Lab 17: Capstone Systems Synthesis

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you should be able to:

1. Connect cell structure, genetics, evolution, and ecology in one causal explanation.
2. Trace how molecular changes can scale up to organism, population, and ecosystem effects.
3. Use evidence from multiple course units to support a biological claim.
4. Identify feedback loops and tradeoffs in a biological system.
5. Draw a capstone systems blueprint that connects at least four levels of biology with labeled flows of matter, energy, and information, including at least one feedback loop, and defend it against peer critique.

Overview

This capstone lab asks you to build a systems explanation instead of memorizing separate units. You will follow one scenario from molecule to ecosystem and show how the course ideas fit together.

Materials

- Pencil or pen
- Two sheets of blank paper (one for your blueprint draft, one for a partner's swap review)
- This worksheet

Scenario

Use a fictional ecosystem to test a proposed explanation. A drought reduces water availability. Some plants already carry a heritable variant that may improve water-use efficiency, but may reduce growth when water is plentiful. Predict how plants, herbivores, predators, and decomposers might respond. These are **hypotheses**, not supplied results; identify the evidence needed for each link.

Part 1: Scale Map

Complete the table with one effect at each scale.

Scale	What changes in the scenario?	Evidence or reasoning
Molecule / gene		
Cell		
Organism		
Population		
Community		
Ecosystem		

Part 2: Causal Chain

1. Write a six-step causal chain from genetic variation to ecosystem-level change.

2. Where could natural selection act in this scenario? Identify the trait, environment, and fitness difference.

3. Where could feedback occur? Describe one reinforcing or balancing feedback loop.

Part 3: Evidence Check

Choose at least three course modules from Modules 01–16 and explain how each helps you reason about the drought scenario. Include one evolutionary idea and one ecological idea.

Module	Concept used	How it supports your explanation

Part 4: Systems Blueprint

4. Plan before you draw. Before any drawing, list the **at least four levels** you will connect (choose from: molecule, organelle, cell, tissue, organism, population, community, ecosystem) and describe how a change in one node affects the next. Where an arrow represents a transfer, identify what flows (matter, energy, or information) and in which direction.

5. Draw your blueprint. Now draw **one causal loop diagram** that matches your plan. Requirements:

- At least **four levels** from the list above, each written as a labeled node (a simple circle or box).
- Label each causal arrow with a mechanism (for example, "more crowding reduces food per plant"). Where an arrow represents a transfer, also label **matter**, **energy**, or **information**. A causal influence need not transfer a substance.
- At least **one feedback loop**: a closed causal chain. Mark each arrow + when increasing the first quantity increases the next, or – when it decreases the next, with other conditions held constant. Label the whole loop **reinforcing** if it amplifies change or **balancing** if it counteracts change. For example, plant density → competition → per-capita reproduction → plant density forms a balancing loop.

6. Rank the proposed tradeoffs. Rank these hypotheses **1–3** for relevance to the drought scenario (1 = most relevant). Explain what measurement would test your top choice; do not assume every cost occurs in this ecosystem.

Tradeoff	Rank (1–3)	One-line justification
Water conservation vs. carbon dioxide uptake		
Reproduction now vs. survival later (organisms)		
Root investment vs. leaf growth		

7. Peer defense. Swap blueprints with a partner. Find the **weakest arrow** in their loop — the link you find least convincing — and either defend or attack it in **two sentences** (write about *their* diagram; then they do the same for yours).

Capstone Claim

8. Make one claim about the drought scenario using at least three ideas from different BIOL-1 modules. Name an **alternative explanation**, a **measurement or comparison** that distinguishes the explanations, and a **result that would make you revise your claim**. Separate your predictions from observations you actually have.